Atticus Rothschild

Mechanical engineering | Rice University

B.S. Mechanical Engineering - Expected 2029

I'm a mechanical engineering student at Rice University, interested in semiconductor equipment and manufacturing.

At EMERGE Lab, I fabricate transistors in the cleanroom for research into predicting device behavior. Previously, at Kong Research Group, I worked on printed electronics and thermal experiments.

I also lead Rice Racing's powertrain subteam.

My personal projects include loudspeaker design and amateur radio.

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Engineering portfolio | October 2026

EMERGE Lab

At Rice's EMERGE Lab, I fabricate transistors in the cleanroom for research into predicting device behavior, with planned work on microelectromechanical switches.

Transistor fabrication

I'm responsible for fabricating transistors for a project exploring whether machine-learning models can predict the behavior of other transistors made from the same material. My role is device fabrication; a collaborator will handle electrical testing and model training. The devices have not yet been electrically tested.

We're starting with silicon, with plans to extend the work to two-dimensional materials and PEDOT-based bioelectronic devices for neural interfaces. The PEDOT devices will initially be supplied by another lab; fabricating them ourselves is a possible later step. This part of the project will explore predictive models for devices made from a conducting polymer used in bioelectronics.

Cleanroom processes

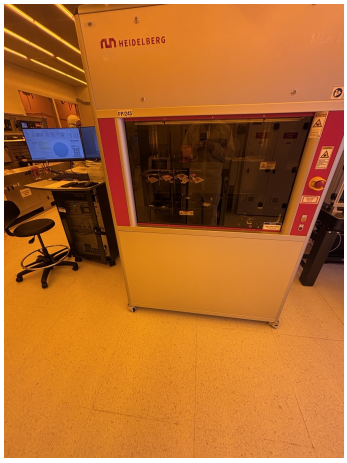
My hands-on work includes spin-coating photoresists, baking, photolithography and development, chemical removal of polymer layers, and plasma cleaning. I also adjust process recipes, including spin speed and bake time, to suit the equipment and achieve the required samples.

I also use atomic layer deposition (ALD), ellipsometry, and wafer dicing as part of the fabrication workflow. These processes span thin-film deposition, film characterization, and separating wafers into individual chips.

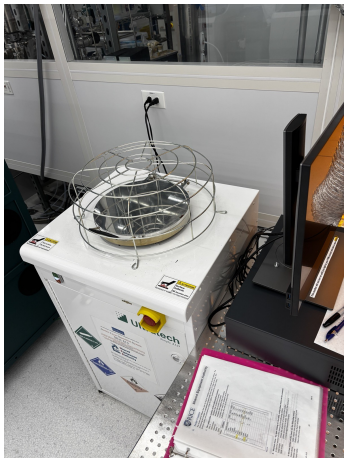
Improving coating quality

During spin-coating, I encountered comet-shaped streaks and an uneven surface. I traced the issue to my dispensing technique introducing bubbles into the photoresist.

I slowed the dispensing process and used a little extra resist to avoid introducing air. Subsequent coatings showed fewer comets and a smoother surface.



Photolithography equipment.



Atomic layer deposition system.

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EMERGE Lab | Design & positioning

MEMS microrelay

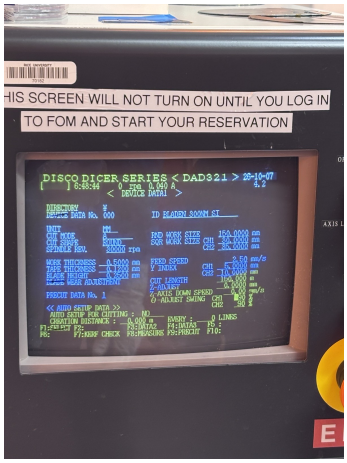
A planned extension of the transistor project is an electrostatically actuated MEMS microrelay, drawing on published relay research coauthored by Elad Alon. An applied voltage creates an electrostatic force that deflects a spring structure, using mechanical motion to open or close an electrical connection.

The research goal is to explore whether the same modeling approach can predict the relay's behavior, extending the investigation from electronic transistors to mechanical switching devices. This work is at the planning stage.

Laser testing & positioning

My earlier work focused on a micromanipulator and light-tight enclosure for laser absorption testing of two-dimensional-material chips. The setup supports repeatable sample positioning while controlling light entering the experiment.

I designed the full mechanical setup in CAD, including the light box, chip holder, and positioning hardware. I assembled the micromanipulator using mostly 3D-printed parts, purchased motors, and electronics. The light box and chip holder remain at the CAD stage. The positioning system uses a Raspberry Pi Pico, stepper motors, and encoders to support sample alignment for optical measurements.



Wafer dicing setup.

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Kong Research Group

I investigated how material formulation, printing conditions, and layer thickness affect the fabrication and thermal response of printed electronics on heat-sensitive surfaces.

Protecting biological surfaces

I investigated printable silicone layers intended to protect leaf surfaces during microplasma processing of conductive silver ink. I prepared PDMS and Ecoflex formulations, adjusted fumed silica content for printability, and explored thermally expanding microspheres as a foaming agent.

The challenge was to produce a consistent insulating surface while limiting heat damage to the substrate beneath it.

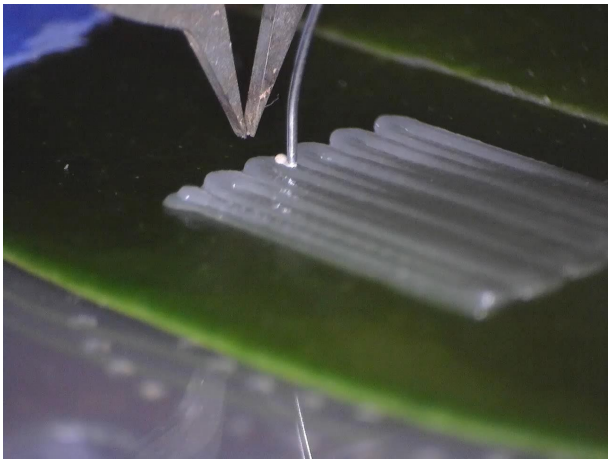
Print-path development

I created and refined DXF printing paths in SolidWorks and worked with the Python workflow that converts them into printer instructions. I varied line spacing, nozzle height, and extrusion conditions to improve layer uniformity and produce samples for comparison.

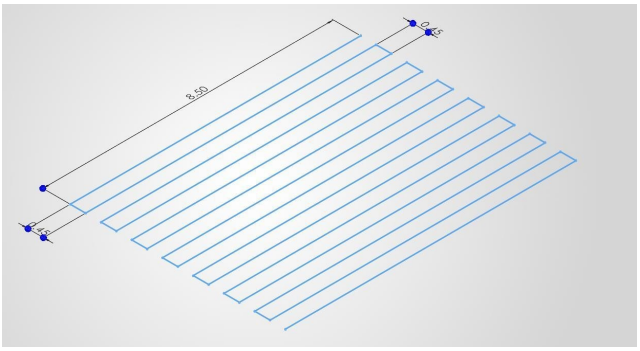
Sensor fabrication & testing

I printed silver traces, created insulating bridges with Ecoflex, and added conductive connections across them. I checked continuity with a multimeter and tested a fabricated sensor using a vector network analyzer, examining its resonance and coupling behavior.

I also worked on carbon and silver circuits on thermoplastic shrink film, including resistive heating patterns for a drone-folding project. Resistance variation and surface warping were ongoing fabrication challenges.



Printing on biological surfaces.



Print-path development.

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Kong Research Group | Experiments

Thermal experiments

I prepared samples with and without foaming agent, measured their thickness under a microscope, and used an IR camera to record their response to repeated microplasma pulses. The setup included a custom sample holder for underside temperature measurements.

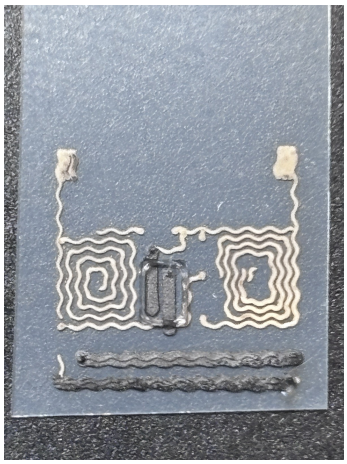
I refined the experiment plan to extend the measured thickness range and establish a non-foaming baseline. Microscope measurements showed that the finished layer thickness did not simply equal the programmed nozzle height, so I used measured thickness when comparing samples.

During testing, I kept probe spacing and pulse timing consistent, positioned silver drops near the sample centers, and used a dry-air supply to help control humidity. I extracted peak-temperature data from the recordings and used Python plots to compare thickness and material conditions.

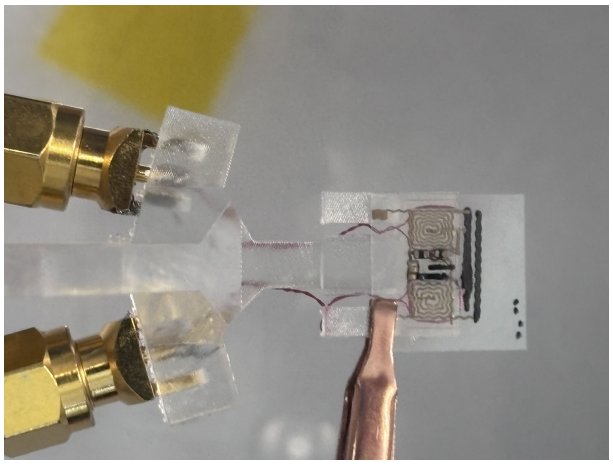
Results

In these trials, sample thickness showed a much stronger relationship with measured temperature than the foaming-agent condition. The initial foaming-agent comparison was inconclusive; I subsequently prepared a higher-concentration mixture for further testing.

This made thickness control a central part of the experiment, rather than treating nozzle height alone as a measure of the finished layer.



Printed sample.



Sample connections for testing.

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Rice Racing | Cooling

I lead Rice Racing's powertrain subteam as we develop our first Formula SAE car around a Honda CBR600RR engine. My work spans cooling calculations, fuel-system sizing, component selection, and fitting those systems into the car.

Cooling calculations

With no previous car data to size from, I worked through engine energy balances, estimated coolant heat loads, and comparable Formula SAE designs. I then used coolant flow, airflow, and inlet temperatures to examine the heat transfer required from the radiator.

The current sizing study uses a provisional 21.8 kW heat-rejection target. I evaluated how changes in core dimensions, coolant flow, and airflow affect predicted capacity, using a Python heat-exchanger model alongside ANSYS Fluent airflow studies.

This work showed why the radiator dimensions alone are not enough: the core, fans, and duct need to work together. The current package remains a design candidate, with supplier performance data and testing needed to establish its cooling capacity.

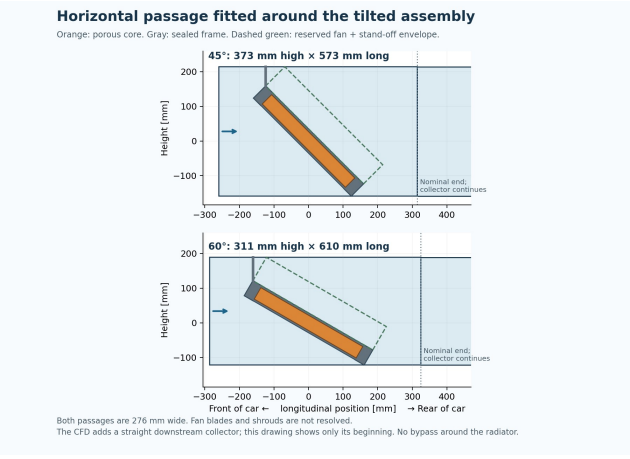
Fans, ducting & packaging

I compared single- and dual-puller fan arrangements using pressure-flow curves, estimated radiator resistance, and electrical current demand. This connects the fan choice to the airflow it can deliver through the installed system, including the shroud and outlet.

I'm also comparing 45° and 60° radiator layouts to reduce sidepod height while allowing space for the fans, hoses, and discharge airflow. Supplier discussions cover single-row core construction, overall tank dimensions, and manufacturability.



Engine on the workshop bench.



Preliminary radiator packaging study.

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Rice Racing | Fuel systems

Fuel consumption & tank sizing

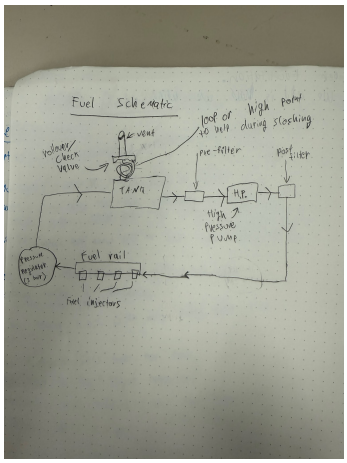
I'm evaluating how much usable fuel the car needs for endurance, balancing capacity, weight, and packaging. The work includes preliminary Realis WAVE engine simulations at different engine speeds and throttle openings, incorporating the team's intake and exhaust dimensions.

I used those fuel-flow results to compare driving-load scenarios and allowances for starting, idling, reserve fuel, and fuel the pickup cannot reach. The driving-time assumptions still need validation; the simulations help assess tank-size choices but do not yet establish a measured endurance requirement.

Fuel delivery & integration

The fuel-system work also covers pump and regulator selection, filtration, rail connections, and hose routing. I'm evaluating a HydraMat pickup with a tank footprint constrained by the pickup, making tank height an important capacity decision.

I've worked through the distinction between internal tank volume, fuel at the competition fill level, and fuel that remains usable during driving. Reviewing the team's schematics, parts lists, and engine inputs has also helped identify details that need to agree before fabrication, including the pump assembly and injector configuration.



Early fuel-system concept; component choices have evolved.

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Rapyuta Robotics | Simulation

I worked on the Unreal Engine simulation for Rapyuta ASRS, an automated storage and retrieval system, during its pre-release development. My work connected CAD, warehouse layout logic, and testing with physical robots.

Picking-station simulation

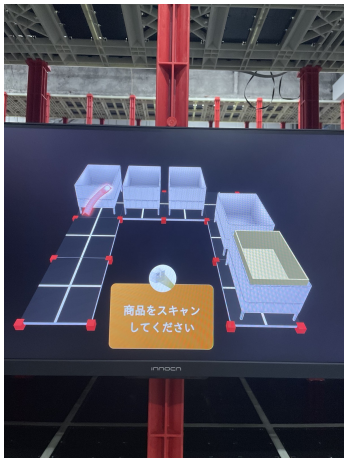
I modeled the existing human picking-station layout in CAD and integrated it into the simulation. In the ASRS workflow, robots bring storage bins to a station where an operator picks items.

I developed the placement logic using a combination of Unreal Blueprints and C++, integrated with ROS. The simulation also reproduced robot communication through ROS nodes, representing how the robots exchanged information as well as how they moved. The logic used the relative positions and orientations of warehouse tiles to identify a U-shaped arrangement and determine where the station should be generated. This linked the station's placement to the surrounding layout.

Adding the picking station made the simulated environment more representative of the physical system and supported testing software and hardware changes before trying them on the real equipment. It also provided a foundation for future simulations of robotic arms performing the picking tasks.



Physical ASRS warehouse layout.



Picking-station operator display.

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Rapyuta Robotics | Testing

Testing against physical robots

I tested physical robots against the simulation and investigated why larger groups disconnected or moved slowly. I improved the internal interface used to start robot images and reduced message traffic to address suspected network congestion. Together, these changes helped scale simultaneous physical-robot operation to 300% of its original capacity.

Testing & client demonstrations

I was one of a small group familiar with operating the robots, their hardware, and troubleshooting recurring issues. This included manually resetting bin-lifting pushers that overextended and could no longer retract, and identifying occasions when robots lost line tracking during fast operation.

With that experience, I was trusted to run robot tests with new code and operate real-world and simulated demonstrations for prospective clients.

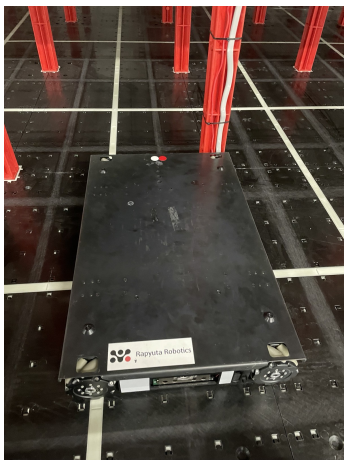
Continuing robotics at Rice

Rice University · MECH 450 · Fall 2026

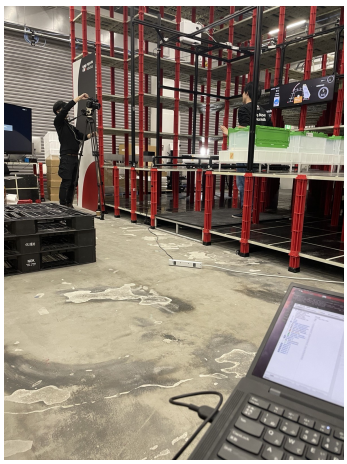
I've continued exploring robotics at Rice through Algorithmic and AI-enabled Robotics (MECH 450), an upper-level course I'm taking as a sophomore with the professor's permission. The course extends my work with robot simulation into the algorithms that plan a robot's movement.

I led the C++ implementation of a random-tree motion planner for a seven-joint Franka Panda robot arm. The planner searches for a route around obstacles by sampling joint configurations, checking connections for collisions, and tracing the resulting path through the tree.

For this team project, my partner built the test environments and ran the trials. Our planner found collision-free paths in two simulated scenes where moving directly from start to goal would hit an obstacle.



Physical ASRS robot.



Demonstration setup.

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Other work | Engineering experience

FIRE1

Medical Device Engineering Work Experience · Dublin · May 2026

Worked with the implantation team to investigate NORM Sensor deployment through a catheter. Parts of the nitinol frame caught on each other as the sensor emerged from the catheter. I ran deployment tests using both benchtop fixtures and anatomical models, varying how the sensor was loaded into the catheter and released to find a method that would prevent the frame from catching on itself. The team had to balance mechanical reliability with how doctors would perform the deployment and what they could see under fluoroscopy (real-time X-ray imaging). Testing was still ongoing when my placement ended.

Engine work

Mechanical Engineering Apprentice · Chatham, NY · 2024-2026

Assisted mechanical engineer Lee Gagne with engine teardown, inspection, reassembly, precision measurement, component refurbishment, and system testing.



Engine work during my apprenticeship with Lee Gagne.

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Other work | Personal projects

Loudspeakers

Adapted an existing loudspeaker design into a custom build, fabricating CNC-routed MDF enclosures and assembling the drivers, passive crossovers, and internal wiring. Evaluated the finished speakers through listening tests and frequency tests using a signal generator. These tests revealed a rattle, which I traced to a loose wire connection. Securing the connection eliminated the rattle on retesting.

The most challenging part was forming the curved cabinet panels. I calculated the kerf spacing and cut depth to create the curve, then carefully handled and shaped the delicate cut panels.

Amateur radio

Amateur Extra licensed operator. Built and tuned multi-band transceivers.



Completed loudspeaker.



Crossover assembly and internal wiring.

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